Jingze Dai

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https://www.linkedin.com/in/jingze-dai/ | https://scholar.google.com/citations?user=o9k4mBkAAAAJ&hl=en

SKILLS AND CAPABILITIES (Front-End | Back-End | Full-Stack | DevOps | CI/CD | QA Testing | MLE)

Technical Skills: **AI/ML Tools** PyTorch/TensorFlow, Scikit-Learn, LangChain, LangGraph, RAG, OpenCV, SciPy, NumPy, CUDA, Pandas, R, JAX, XGBoost, MLFlow; **Software Engineering** Python, Java, C++, C, C#, JavaScript, TypeScript, SDLC, Go, Kotlin, Rust, Swift, Scala, Haskell, Ruby, Perl, React, Next.js, Vue.js, Flask, FastAPI, Express.js, Spring, .NET, HTML, CSS, jQuery, PostgreSQL/MySQL, MongoDB, DynamoDB, Redis, DB2, AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Kafka, GraphQL, Hadoop, Spark, Linux/Unix, Git, Selenium, Android, iOS, Unity, OpenGL, OpenCL, TCP/IP, JSON, XML, Agile.
Certifications: IBM Data Science, Machine Learning Specialization (Coursera), SQL **Advanced** & Angular, Rest APIs, Node.js **Intermediate** Certificate (HackerRank), Salesforce Development Professional, SAS **Advanced**, Deep Learning A-Z (Udemy).

WORK EXPERIENCE

Software Research Assistant — Jan 2024 - Present
Cornell University & McMaster University — Ithaca, NY

- Wrote 6 conditions' stochastic energy systems' simulation models, and conducted independent proofs for each part, which determined the formulas of optimal parameters triggering the minimal two-layer cost. (https://arxiv.org/pdf/2507.03510)
- Devised 20 image-transformed CNN methods to detect V2X misbehaviour messages, reaching a 20-class detection accuracy of 98% and a detection time within 0.06 seconds, using explainable AI methods to select 3 most important driving features.
- Conducted a literature review on Explainable AI's recent advances for IoT security into a survey (**Accepted** at **ISICN 2025**)

Computer Science Course Teaching Assistant (CS 1XC3 & CS 3SH3) — May 2023 - Dec 2023
McMaster University — Hamilton, ON

- Instructed 200+ students in C programming, bash scripts, and LaTeX, guiding the development of practical software projects including a C-based row-cancellation game, 3 synchronization simulators, and 10 mobile programming assignments.
- Reviewed and graded 32 systems programming assignments involving multithreading, concurrent processing, and caching, providing feedback that addressed 110+ technical questions and reinforced student understanding of operating systems.

Software Developer (Co-op) — Sep 2021 - Apr 2022
Government of Ontario, CYSSC services (Ministry of Children, Community, and Social Services) — Toronto, ON

- Developed and deployed 5 public-facing government service web pages (with user profiles) using Angular and REST APIs.
- Designed 1 MySQL relational database (with 3 tables) and 1 MongoDB database, and build 1 backend server with 3 parts of authentication/scheduling APIs. Constructed an Angular tester app simulating human user logins and fingerprint checks.
- Analyzed 1000+ lines of user data, and summarized user appointment's actual childcare needs and time distributions.
- Created 220+ unit/integration tests using PyTest and JUnit, improving reliability and eliminating 18 functional defects.

EDUCATION

Statistics, Cornell University — Ithaca, NY
Master of Professional Studies (M.P.S) — Sep 2025 - Present
Major in Statistics, with specialization in **Data Science, GPA 3.91/4.0**

Computer Science, McMaster University — Hamilton, ON
Bachelor of Applied Science (B.A.Sc), First Class Honor's — Sep 2018 - Apr 2023
Major in Computer Science, Dean's Honor List 2021 – 2023, Graduate with Distinction**, GPA 3.64/4.0**
Relevant Coursework: Algorithms, Data Structures, Machine Learning, Causal Inference, Artificial Intelligence, Natural Language Processing, Web/Mobile Development, Databases, Software Systems, Large Language Models, Compilers, Distributed Computing, Big Data Management, Robotics, Reinforcement Learning, Networking, Security, Computer Visions, Scripting

PROJECTS

MuseScore - Music Notation and Composition (https://musescore.com/) — Apr 2023 - Present

- Contributed to MuseScore Google Summer of Code (GSoC) project, code at https://github.com/musescore/MuseScore.
- Implemented 3 statuses' cursor shape effects when dragging movable panels, and improved panels' merge/detach functions.
- Engineered 2 novel C++/QML approaches to insert braille panels on the view menu, and designed 1 braille panel hotkey.

Drasil - Generate All the Things (https://jacquescarette.github.io/Drasil/) — May 2022 - Present

- Solved 30+ problems of the Drasil research project, supervised by Dr. Jacque Carette and Dr. Spencer Smith at McMaster University. Initial publication available at https://www.cas.mcmaster.ca/~carette/publications/DrasilPositionPaper.pdf.
- Mitigated 25 issues from Haskell automatic generations, re-constructed 32 existing code segments with the expected format.